

I am a PhD student in the final months of my study. I work in theoretical computer science at the boundary of formal logic, universal algebra and computational complexity, drawing on tools from proof theory, algebraic logic, well-quasi-order theory and infinite state systems.

Education

- 2022–present **PhD, Computer Science**, *University of Groningen*, Groningen, The Netherlands
Advisor Prof. Dr. Revantha Ramanayake
- 2020–2022 **MSc, Computer Science**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisors Prof. Dr. João Marcos and Dr. Sérgio Marcelino
Final grade 5.0/5.0
- 2018–2019 **BSc, Computer Science**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisor Prof. Dr. João Marcos
Final grade 9.7/10 summa cum laude
- 2015–2018 **BSc, Information Technology**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisor Dr. Selan Santos
Final grade 9.6/10 summa cum laude
- 2011–2016 **Technician in Informatics**, *Federal Institute of Rio Grande do Norte*, Natal, Brazil
Advisor Dr. Bruno Costa
Final grade 9.7/10 summa cum laude
- 2011–2015 **Technician in Web Programming**, *Federal University of Rio Grande do Norte*, Natal, Brazil

Publications (Logic and Theoretical Computer Science)

Note In computer science, conference publications are peer-reviewed and count as much as journal papers. Moreover, authors are usually listed in alphabetical order.

- [1] A.R. Balasubramanian, Vitor Greati, and Revantha Ramanayake. *Hypersequent calculi have Ackermannian upper bounds*. LICS 2026 (to appear). 2026. arXiv: 2602.19229 [cs.LO].
- [2] Vitor Greati and Revantha Ramanayake. “Tight length theorems for multiset extensions of Higman’s lemma”. In: *Theoretical Computer Science* 1057 (2025), p. 115546. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2025.115546.
- [3] Vitor Greati et al. “Analytic calculi for logics of indicative conditionals”. In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Gian Luca Pozzato and Tarmo Uustalu. Cham: Springer Nature Switzerland, 2025, pp. 59–81. ISBN: 978-3-032-06085-3. DOI: 10.1007/978-3-032-06085-3_4.
- [4] Vitor Greati, Sérgio Marcelino, and Umberto Rivieccio. “Axiomatizing the logic of ordinary discourse”. In: *Information Processing and Management of Uncertainty in Knowledge-*

- Based Systems*. Ed. by Marie-Jeanne Lesot et al. Cham: Springer Nature Switzerland, 2025, pp. 390–405. ISBN: 978-3-031-74000-8. DOI: 10.1007/978-3-031-74000-8_32.
- [5] Vitor Greati et al. “Generating proof systems for three-valued propositional logics”. In: *Handbook of Three-Valued Logics (to appear)*. 2025. arXiv: 2401.03274 [math.LO].
 - [6] Vitor Greati, Sérgio Marcelino, and Umberto Riveccio. “Finite Hilbert systems for Weak Kleene logics”. In: *Studia Logica* 112.6 (2024), pp. 1215–1241. ISSN: 1572-8730. DOI: 10.1007/s11225-023-10079-w.
 - [7] Vitor Greati et al. “Adding an implication to logics of perfect parafinite algebras”. In: *Mathematical Structures in Computer Science* 34.10 (2024), 1138–1183. DOI: 10.1017/S0960129524000227.
 - [8] Vitor Greati and Revantha Ramanayake. “Deducibility in the full Lambek calculus with weakening is H_{ACK}-complete”. In: *Advances in Modal Logic, AiML 2024, Prague, Czech Republic, August 19-23, 2024*. Ed. by Agata Ciabattoni, David Gabelaia, and Igor Sedlár. College Publications, 2024, pp. 401–422. URL: www.aiml.net/volumes/volume15/19-GreatiRamanayake.pdf.
 - [9] Vitor Greati. “Hilbert-Style formalism for two-dimensional notions of consequence”. MA thesis. Universidade Federal do Rio Grande do Norte, 2022. URL: <https://repositorio.ufrn.br/handle/123456789/46792>.
 - [10] Vitor Greati and João Marcos. “Finite two-dimensional proof systems for non-finitely axiomatizable logics”. In: *Automated Reasoning*. Ed. by Jasmin Blanchette, Laura Kovács, and Dirk Pattinson. Cham: Springer International Publishing, 2022, pp. 640–658. ISBN: 978-3-031-10769-6. DOI: 10.1007/978-3-031-10769-6_37.
 - [11] Vitor Greati, Sérgio Marcelino, and João Marcos. “Proof search on bilateralist judgments over non-deterministic semantics”. In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Springer International Publishing, 2021, pp. 129–146. ISBN: 978-3-030-86059-2. DOI: 10.1007/978-3-030-86059-2_8.
 - [12] Joel Gomes et al. “On logics of perfect parafinite algebras”. In: *Proceedings 16th Logical and Semantic Frameworks with Applications, LSFA 2021, Buenos Aires, Argentina (Online), 23rd - 24th July, 2021*. Ed. by Mauricio Ayala-Rincón and Eduardo Bonelli. Vol. 357. EPTCS. 2021, pp. 56–76. DOI: 10.4204/EPTCS.357.5.
 - [13] Vitor Greati. “Hilbert calculi for the main fragments of Classical Logic”. Undergraduate Thesis. Federal University of Rio Grande do Norte, 2019. URL: <https://repositorio.ufrn.br/items/f737862d-3ced-4c8d-bf7b-e7913c2246be>.

Preprints

- [14] Amirhossein Akbar Tabatabai, Vitor Greati, and Revantha Ramanayake. *Predicative ordinal recursion on the constructive Veblen hierarchy*. Submitted to APAL. 2025. arXiv: 2510.18497 [math.LO].
- [15] Nikolaos Galatos et al. *Complexities of well-quasi-ordered substructural logics*. 2025. arXiv: 2504.21674 [cs.LO].

Submitted

- [16] Vitor Greati and Revantha Ramanayake. *Commutative full Lambek calculus with contraction and Kleene star is Π_1^0 -complete*. Submitted to LICS 2026. 2026.
- [17] Nikolaos Galatos et al. *Complexity bounds for commutative knotted substructural logics via proof theory*. Submitted to APAL. 2025.

- [18] Nikolaos Galatos et al. *Lower complexity bounds for knotted substructural logics via counter machines*. Submitted to Transactions of the American Mathematical Society. 2025.

Publications (Data Science, Computer Vision, Material Sciences)

- [19] Gilles Silvano et al. "Synthetic image generation for training deep learning-based automated license plate recognition systems on the Brazilian Mercosur standard". In: *Design Automation for Embedded Systems* 25.2 (2021), pp. 113–133. ISSN: 1572-8080. DOI: 10.1007/s10617-020-09241-7.
- [20] Gilles Velleneuve Trindade Silvano et al. "Artificial Mercosur license plates dataset". In: *Data in Brief* 33 (2020), p. 106554. ISSN: 2352-3409. DOI: 10.1016/j.dib.2020.106554.
- [21] André Fiel Borges et al. "Wearable optical device for the visually impaired in the classroom". In: *Research, Society and Development* 9.9 (2020). DOI: 10.33448/rsd-v9i9.7623.
- [22] Roberto R. C. Lima et al. "Sodium-modified vermiculite for calcium ion removal from aqueous solution". In: *Industrial & Engineering Chemistry Research* 58.22 (2019), pp. 9380–9389. DOI: 10.1021/acs.iecr.9b00045.
- [23] Aguinaldo Bezerra et al. "Enabling interactive visualizations in industrial big data". In: *IFAC* 53.2 (2020). 21st IFAC World Congress, pp. 11162–11167. ISSN: 2405-8963. DOI: 10.1016/j.ifacol.2020.12.292.
- [24] Vinícius Ribeiro et al. "Brazilian Mercosur license plate detection: a deep learning approach relying on synthetic imagery". In: *2019 IX Brazilian Symposium on Computing Systems Engineering (SBESC)*. 2019, pp. 1–8. DOI: 10.1109/SBESC49506.2019.9046091.
- [25] Aguinaldo Junior et al. "An industrial big data processing engine". In: *14^o Simpósio Brasileiro de Automação Inteligente*. 2019. DOI: 10.17648/sbai-2019-111445.
- [26] Vitor Greati et al. "Monitoring platform for vehicle irregularities in Smart Cities". In: *Congresso Brasileiro de Automação*. 2018 (in Portuguese).
- [27] Vitor Greati et al. "Method for license plate recognition in uncontrolled environments". In: *Simpósio Brasileiro de Automação Industrial*. 2017 (in Portuguese).
- [28] Vitor Greati, Vinícius Campos, and Ivanovitch Silva. "System based on automatic license plate recognition for vehicle monitoring and characterization of patrons in closed environments". In: *Escola Potiguar de Computação e suas Aplicações*. 2017 (in Portuguese).
- [29] Vitor Greati et al. "A Brazilian license plate recognition method for applications in smart cities". In: *2017 IEEE First Summer School on Smart Cities (S3C)*. 2017, pp. 43–48. DOI: 10.1109/S3C.2017.8501395.
- [30] Bruno Sielly Jales Costa et al. "A visual protocol for autonomous landing of unmanned aerial vehicles based on fuzzy matching and evolving clustering". In: *2015 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. 2015, pp. 1–6. DOI: 10.1109/FUZZ-IEEE.2015.7337907.

Presentations

Invited talks

- 2025 Automated generation of Hilbert calculi for finite-valued logics. *Workshop Kleene logics and beyond*, Cagliari, Italy.
- 2022 Finite analytic two-dimensional proof systems for non-finitely axiomatizable logics. *Local seminar at the University of Bern*, Bern, Switzerland.

Conferences

- 2025 Analytic calculi for logics of indicative conditionals. *TABLEAUX'25*, Reykjavík, Iceland.
- 2025 Predicative recursion on the Veblen hierarchy. *Proof Society Workshop*, Ghent, Belgium.
- 2024 Deducibility in the full Lambek calculus with weakening is HAcK-complete. *AiML'24*, Prague, Czech Republic.
- 2024 Axiomatizing the logic of ordinary discourse. *IPMU 2024*, Lisbon, Portugal.
- 2024 Knotted substructural logics have fast-growing complexity. *Logic Colloquium 2024*, Gothenburg, Sweden.
- 2023 Decidability and complexity upper bounds for weakly commutative knotted substructural logics. *MOSAIC 2023*, Vienna, Austria.
- 2023 Hilbert-style formalism for two-dimensional notions of consequence. *VCLA Awards*, Vienna, Austria.
- 2022 Finite analytic two-dimensional proof systems for non-finitely axiomatizable logics. *IJCAR 2022*, Haifa, Israel.
- 2022 Finite and analytic Hilbert-style systems for Paraconsistent Weak Kleene and Bochvar-Kleene logics. *Trends in Logic XXII*, online.
- 2021 Proof search on bilateralist judgments over non-deterministic semantics. *TABLEAUX 2021*, online.
- 2019 Towards the automated extraction of Hilbert calculi for fragments of classical logic. *19th Brazilian Logic Conference*, Joao Pessoa, Brazil.
- 2018 Platform for monitoring irregularities in vehicles in Smart Cities. *Congresso Brasileiro de Automática (CBA)*, Joao Pessoa, Brazil.
- 2017 A Brazilian license plate recognition method for applications in smart cities. *1st IEEE Summer School on Smart Cities*, Natal, Brazil.
- 2015 A visual protocol for autonomous landing of unmanned aerial vehicles based on fuzzy matching and evolving clustering. *FUZZ-IEEE 2015*, Istanbul, Turkey.

Seminars

- 2026 Fast-growing complexities in substructural logics. *PhD seasonal seminar*, Groningen, The Netherlands.
- 2024 Deducibility in the full Lambek calculus with weakening is HAcK-complete. *NetTCS*, Leiden, The Netherlands.
- 2024 Computational properties of non-classical logics. *AILO talk*, Groningen, The Netherlands.

Round tables

- 2025 Computational tools for logic and algebra, *Workshop Kleene logics and beyond*, Cagliari, Italy.

Teaching and dissemination

Teaching

2023-2026 **Computational complexity**, BSc course

- Description: Teaching assistant and lecturer (tutorials, assignments, grading, some lectures).
- Institution: Faculty of Science and Engineering, University of Groningen.
- When: Fall 2023, Fall 2024, Fall 2025.

- 2023–2026 **Algorithmic programming contests**, BSc course
- Description: Contest assistant.
 - Institution: Faculty of Science and Engineering, University of Groningen.
 - When: 2023, 2024, 2025.
- 2021–2021 **Mathematics for Computer Science III**, BSc course
- Description: Teaching assistant.
 - Institution: Computer Science Department (DIMAP), UFRN.
- 2014–2015 **Programming contests in high school**, Institutional project
- Description: Tutor in the project “Training high school students for algorithm competitions”, funded by CNPq-MEC.
 - Institution: IFRN.
- 2012–2012 **Web Programming**, Technical course
- Description: Teaching assistant.
 - Institution: Instituto Metropole Digital, UFRN
- Dissemination**
- 2019 **Fundamentals of Python and its applications in Chemistry**, Short course (4h), II Chemistry Week, UFRN
- 2019 **Breaking vulnerable CAPTCHAS with image processing techniques**, Short course (4h), Campus Party Natal 2
- 2016, 2017 **Introduction to L^AT_EX**, Short course (8h), Science and Technology Week, IFRN

Professional service

Supervision

- 2025 **Master’s research internship**, *University of Groningen*
- Description: Co-supervisor of the MSc student Wouter Leferink in his internship program.
 - Project: On a combinatorial proof for upper and lower bounds of bad sequences over $\mathbb{N}^k$.
- 2025 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Scott Lantinga.
 - Project: An interactive tool for exploring rule permutations in structural proof theory.
- 2024 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Bogdan Bruma.
 - Project: Boolean circuits in computational complexity: an accessible account (survey).
- 2023 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Wouter Leferink.
 - Project: Complexity classes beyond elementary (survey).
- 2023 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Alexandra Stan.
 - Project: Upper bounds for substructural logics using well-quasi-orders (survey).

Organization

- 2023 Head of the jury of the Groningen Algorithmic Programming Contest (GAPC).
- 2018 IEEE Symposium on Computers and Communications.
- 2016 IEEE Conference on Evolving and Adaptive Intelligent Systems.
- 2014 Science and Technology Week, IFRN.

Reviewer for journals

- 2025 International Journal of Approximate Reasoning (Elsevier)

- 2024 Logic Journal of the IGPL (Oxford Academy)
- 2024 Journal of Applied Non-Classical Logics (Taylor and Francis)
- 2024 Order (Springer)
- 2022 Journal of Logic and Computation (Oxford Academy)
- 2021 Studia Logica (Springer)
- 2021 Logic and Logical Philosophy (Nicolaus Copernicus University)

Member of assessment committees

- 2022 **Bachelor thesis defense**, *Department of Computer Engineering and Automation*, UFRN
 ○ Description: Member of the assessment committee of the BSc defense of Mateus Abrantes.
- 2018 **Project assessment**, *V Exposition of Science and Technology of the North of Natal*, IFRN
 ○ Description: Member of the assessment committee of research projects.

Experience in industry

- 2020–2022 **Researcher and developer**, *Logap Systems*, Natal, Brazil
 ○ Description: Research leader and developer of the windfarm management system *Windbox*.
- 2015–2016 **Web Development Intern**, *IT Department*, UFRN
 ○ Description: Web developer of the system SIGAA, used by the university as the main academic management system.

Grants and honours

Grants

- 2023 **E.W. Beth foundation**, *KNAW WF/2463-23*, Visits from external researchers and a three-day symposium on computational properties of substructural logics.

Honours

- 2023 **VCLA Outstanding Master Thesis Award – Runner-up**, *Vienna Center for Logic and Algorithms*, Vienna, Austria
- 2019 **2nd Best Paper Award**, *IX Brazilian Symposium on Computing Systems Engineering*, Natal, Brazil
- 2014 **Silver Medal**, *CRIA's Algorithm Cup*, Rio Info, Rio de Janeiro, Brazil
- 2014 **Bronze Medal**, *10th OBMEP*, IMPA, Brazil
- 2014 **Third best project of Natural Sciences and Mathematics**, *MOCITEC*, Federal Institute of Rio Grande do Norte, Natal, Brazil
- 2014 **Outstanding research award**, *MOCITEC IFRN*, ABRITEC, Natal, Brazil
- 2014 **Outstanding student award**, *Federal Institute of Rio Grande do Norte*, Natal, Brazil

Skills and competences

Languages

Portuguese

Native

English

Fluent

Programming languages

■■■■■ C++, Java, Python,
 $\text{\LaTeX}$

■■■■■ Haskell, Scala



Lean, PHP

Software engineering, data science and others



Git

CLI, GitHub, GitLab, CI/CD



Linux

Debian-based, Arch-based



Docker



Web frontend

HTML5, CSS3, Javascript



Web backend in Java

Spring, Hibernate



Web backend in Python

django



Web backend in PHP

PHP Ignite



Data science in Python

pandas, scikit-learn, dask



Databases

MySQL, PostgreSQL, MongoDB



Desktop programming

Java



Android programming

Java

Additional information

2020-2022 Student member of the Association for Symbolic Logic (ASL).

2019-2020 Volunteer in IEEE Computer Society at UFRN.